

A SIMPLE GROUP WITH FULL MF RADICAL

ANONYMOUS

A countable group is MF if it embeds in the unitary group of a norm matrix corona. We construct a nontrivial finitely generated simple Kazhdan group H such that every homomorphism from H to an MF group is trivial. Equivalently, the MF radical of H is all of H , so H is non-MF. We take

$$H = \mathrm{EL}_{16}(L_{\mathbb{F}_2}(1, 2)).$$

Conjugation by an explicit Whitehead matrix maps a Kazhdan subgroup $L \cong \mathrm{EL}_4(L_{\mathbb{F}_2}(1, 2))$ into itself. If c centralizes L , one-sided Kazhdan transport implies that the commutators $[ucu^{-1}, \ell]$ converge to the identity in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation. We prove that a normal Kazhdan subgroup generated by such commutators lies in the MF radical. For our choice of u, c , and ℓ , the commutator is an elementary matrix that normally generates H . Every finite-dimensional linear representation of H over any field is also trivial. The same group yields a general construction: for every countable group Q , there is a split epimorphism $\pi_Q: W_Q \rightarrow Q$ such that

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

If Q is MF, the kernel of π_Q is the MF radical of W_Q and is normally generated by the single element d above.

1. INTRODUCTION

A countable group is *MF* if it embeds in the unitary group of a norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Equivalently, it admits finite-dimensional unitary models whose multiplicative defects tend to zero in operator norm and which asymptotically separate its nonidentity elements. This is the convention of Carrión–Dadarlat–Eckhardt [CDE].

For a countable group G , define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

The group G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

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The existence of a countable non-MF group was open [Sh]. Theorem A answers this question and proves the stronger equality $\text{Rad}_{\text{MF}}(H) = H$ for an explicit simple group.

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(1) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem A. *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{16}(R)$. Then H is nontrivial, simple, finitely generated, and has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF.

Call an element of G *universally Hilbert–Schmidt trivial* if its value converges to the identity in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation of G .

Theorem B (normal Kazhdan subgroups). *Let G be countable, let $D \leq G$ be a subgroup of universally Hilbert–Schmidt trivial elements, and let $K \trianglelefteq G$ have property (T). If $K \leq D$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

Suppose that $L \leq G$ has property (T), that $u \in G$ satisfies $uLu^{-1} \leq L$, and that c centralizes L . For every operator-norm asymptotic representation (V_n) of G ,

$$(2) \quad \left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \rightarrow 0 \quad (\ell \in L).$$

Inside H , an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the upper-left subgroup $L \cong \text{EL}_4(R)$. The element $c = e_{45}(1)$ centralizes L , and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

Two elementary commutator identities imply that d normally generates H . The universally Hilbert–Schmidt trivial elements form a normal subgroup, so every element of H has this property. Theorem B then gives $\text{Rad}_{\text{MF}}(H) = H$.

For an exact finite-dimensional representation, conjugation by a compressor defines an injective endomorphism of the commutant of L . This endomorphism is surjective because the commutant is finite-dimensional. Consequently, every finite-dimensional linear representation of H over every field is trivial.

The group H also gives non-MF groups with prescribed MF quotients. The following theorem is proved in Section 7.

Theorem C (prescribed MF quotients). *For every countable group Q , there are a countable group W_Q and a split epimorphism $\pi_Q: W_Q \rightarrow Q$ such that*

$$\text{Rad}_{\text{MF}}(W_Q) = \pi_Q^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\text{Rad}_{\text{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q},$$

where d is the element in Proposition 6.2. Moreover, precomposition with π_Q induces a bijection

$$\text{Hom}(Q, M) \longrightarrow \text{Hom}(W_Q, M)$$

for every MF group M .

Relation to prior work. Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations yield the internal endomorphism of the elementary matrix group used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser's normal-subgroup theorem for elementary groups over exchange rings implies that H is simple [Pre]. The theorem of Ershov and Jaikin-Zapirain gives property (T) for elementary groups over finitely generated rings [EJZ]; standard background on property (T) may be found in [BHV]. Carrión, Dadarlat, and Eckhardt relate the MF property for groups to quasidiagonality of group C^* -algebras [CDE], and Shulman studies permanence of the MF property under amalgamated free products [Sh]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT]; Dadarlat studies obstructions to matricial stability in [Dad].

2. ONE-SIDED COMPRESSION IN FINITE DIMENSION

Theorem 2.1 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k -vector space, and $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\text{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. It commutes with every $\rho(\ell)$, $\ell \in L$, which proves the last assertion. $\square$

Let $\text{Comp}_G(L)$ consist of the $u \in G$ satisfying $uLu^{-1} \leq L$, and let $G^+(L)$ be the subgroup they generate. Define

$$(3) \quad D(G, L) = \langle [gcg^{-1}, \ell] : g \in G^+(L), c \in C_G(L), \ell \in L \rangle_G.$$

Corollary 2.2. *Every finite-dimensional linear representation of G , over every field, is trivial on $D(G, L)$. In particular, if $D(G, L) = G$, then every such representation of G is trivial.*

Proof. For a fixed representation ρ , the elements $g \in G$ for which $\rho(g)C\rho(g)^{-1} = C$, where $C = \rho(L)'$, form a subgroup. Theorem 2.1 shows that this subgroup contains $\text{Comp}_G(L)$ and therefore $G^+(L)$. Every commutator in (3) lies in $\ker \rho$, so its normal closure does as well. $\square$

3. KAZHDAN TRANSPORT IN NORMALIZED HILBERT–SCHMIDT NORM

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator-norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator-norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G . Define

$$(4) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator-norm asymptotic representations of G . An element is universally Hilbert–Schmidt trivial precisely when it belongs to $R_{\infty \rightarrow 2}(G)$.

Lemma 3.1. *For every sequence (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v . Then $x_n^*x_n \rightarrow 1$ in norm. For all sufficiently large n , the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. In a stably finite unital C^* -algebra, equivalent projections $p \leq q$ satisfy $p = q$. $\square$

Theorem 3.2 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator-norm asymptotic representation of G , and let (x_n) be uniformly bounded in operator norm. If*

$$\|V_n(\ell)x_nV_n(\ell)^* - x_n\|_2 \rightarrow 0 \quad (\ell \in L),$$

then, for every $\ell \in L$,

$$\begin{aligned} \|V_n(\ell)V_n(u)x_nV_n(u)^*V_n(\ell)^* - V_n(u)x_nV_n(u)^*\|_2 &\rightarrow 0, \\ \|V_n(\ell)V_n(u)^*x_nV_n(u)V_n(\ell)^* - V_n(u)^*x_nV_n(u)\|_2 &\rightarrow 0. \end{aligned}$$

Proof. Let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the Hilbert-space ultraproduct $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

Its action on $\mathcal{K}_\omega$ is $\sigma(g)$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection of L under $\tilde{\sigma}|_L$. On $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$ [BHV, Chapter 3]. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies

$$U^*PU \leq P.$$

The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$. Hence both U and U^* preserve $\text{Fix } \sigma(L)$, so $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. The two limits in the statement therefore vanish along ω . Since this holds for every free ultrafilter, both sequences converge ordinarily. $\square$

Corollary 3.3. *Let $L \leq G$ have property (T). Then*

$$D(G, L) \leq R_{\infty \rightarrow 2}(G).$$

Proof. Let (V_n) be an operator-norm asymptotic representation. If $c \in C_G(L)$, the sequence $(V_n(c))$ asymptotically commutes with $V_n(L)$ in operator norm and hence in normalized Hilbert–Schmidt norm. Repeated application of Theorem 3.2 shows that the same is true of $(V_n(gcg^{-1}))$ for every $g \in G^+(L)$. Therefore

$$\|V_n([gcg^{-1}, \ell]) - 1\|_2 \rightarrow 0 \quad (\ell \in L).$$

The subgroup $K_2(V)$ is normal, so it contains the normal closure of these elements. Intersecting over (V_n) proves the claim. $\square$

4. NORMAL KAZHDAN SUBGROUPS AND THE MF RADICAL

Lemma 4.1 (asymptotically invariant corners). *Let (V_n) be an operator-norm asymptotic representation of G , and let $q_n \in M_{d_n}(\mathbb{C})$ be nonzero projections such that*

$$\|[q_n, V_n(g)]\| \longrightarrow 0 \quad (g \in G).$$

After identifying $q_n M_{d_n}(\mathbb{C}) q_n$ with $M_{\text{rank } q_n}(\mathbb{C})$, there is an operator-norm asymptotic representation (W_n) of G such that

$$\|W_n(g) - q_n V_n(g) q_n\| \longrightarrow 0 \quad (g \in G).$$

Proof. The compression $q_n V_n(g) q_n$ has multiplicative defect tending to zero, and its two Gram operators differ from q_n by $o(1)$ in operator norm. For all sufficiently large n , define $W_n(g)$ to be its polar unitary in the corner. This unitary differs from $q_n V_n(g) q_n$ by $o(1)$. Set the finitely many exceptional coordinates equal to the corner identity. The multiplicative estimates for the compressions then imply those for (W_n) . $\square$

Proof of Theorem B. Suppose that a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_d)$ is nontrivial on K . Set $\bar{G} = \Theta(G)$ and $\bar{K} = \Theta(K)$. Then $\bar{G}$ is a countable MF group, and $\bar{K}$ is a nontrivial normal Kazhdan subgroup of $\bar{G}$. If (W_n) is an operator-norm asymptotic representation of $\bar{G}$, then $(W_n \circ \Theta)$ is an operator-norm asymptotic representation of G . Hence $\Theta(D) \leq R_{\infty \rightarrow 2}(\bar{G})$.

Choose an operator-norm asymptotic representation (V_n) of $\bar{G}$ that separates a fixed nonidentity element of $\bar{K}$ in operator norm. Such a sequence is obtained by lifting the given embedding of $\bar{G}$ into a norm matrix corona and passing to a subsequence. Write $\pi(g) = [V_n(g)]$ for the corresponding homomorphism to that corona.

Choose a finite symmetric Kazhdan generating set $S \subseteq \bar{K}$, with $1 \in S$, and put

$$h = \frac{1}{|S|} \sum_{a \in S} \pi(a).$$

There is $\theta < 1$ such that $\sigma(h) \subseteq [-1, \theta] \cup \{1\}$. The spectral projection

$$P = 1_{\{1\}}(h)$$

is the image of the Kazhdan projection of $\bar{K}$. Put $q = 1 - P$. If $q = 0$, then $P = 1$; the identity $\pi(a)P = P$ for $a \in \bar{K}$ would imply $\pi(a) = 1$ for every $a \in \bar{K}$, contrary to the chosen operator-norm separation. Thus $q \neq 0$.

Every automorphism of $\bar{K}$ fixes its Kazhdan projection. Since $\bar{K}$ is normal in $\bar{G}$, conjugation by $g \in \bar{G}$ restricts to an automorphism of $\bar{K}$. Therefore

$$\pi(g)P\pi(g)^* = P \quad (g \in \bar{G}),$$

and q commutes with $\pi(\bar{G})$. Choose self-adjoint lifts of q and round their spectra at $1/2$. This gives projection lifts $q_n \in M_{d_n}(\mathbb{C})$. After passing to a

diagonal subsequence, we may assume that every q_n is nonzero,

$$\|[q_n, V_n(g)]\| \longrightarrow 0 \quad (g \in \bar{G}),$$

and, for the self-adjoint coordinate averages

$$h_n = \frac{1}{2|S|} \sum_{a \in S} (V_n(a) + V_n(a)^*),$$

that

$$(5) \quad \|(q_n h_n q_n - \theta q_n)_+\| \longrightarrow 0.$$

The commutator estimates follow from the equality of q with its conjugates in the corona. Inequality (5) is the coordinate form of $ghq \leq \theta q$ and follows from continuous functional calculus for the positive part. Since $\bar{G}$ is countable, one diagonal subsequence suffices for all the estimates.

Apply Lemma 4.1 and write W_n for the resulting asymptotic representation on $q_n \mathbb{C}^{d_n}$. Equip this space with its ordinary normalized matrix trace. Taking traces in (5), and using that polar correction changes the compressed matrices by $o(1)$ in operator norm, gives

$$\frac{1}{|S|} \sum_{a \in S} \operatorname{Re} \operatorname{tr}(W_n(a)) \leq \theta + o(1).$$

Consequently, after passing to a subsequence, one fixed $a_0 \in S$ satisfies

$$\|W_n(a_0) - 1\|_2^2 = 2 - 2 \operatorname{Re} \operatorname{tr}(W_n(a_0)) \geq 2(1 - \theta) - o(1).$$

But $a_0 \in \bar{K} \leq \Theta(D) \leq R_{\infty \rightarrow 2}(\bar{G})$, so $\|W_n(a_0) - 1\|_2 \rightarrow 0$. This contradiction proves that Θ is trivial on K . $\square$

Corollary 4.2 (normal Kazhdan compression defect). *Let $L \leq G$ have property (T), and suppose that $K \trianglelefteq G$ has property (T) and satisfies $K \leq D(G, L)$. Then*

$$K \leq \operatorname{Rad}_{\operatorname{MF}}(G).$$

Proof. Corollary 3.3 gives $D(G, L) \leq R_{\infty \rightarrow 2}(G)$, and Theorem B applies with $D = R_{\infty \rightarrow 2}(G)$. $\square$

5. THE BINARY LEAVITT SELF-COMPRESSION

Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0 t_0, \quad q = s_1 t_1.$$

The relations (1) give

$$(6) \quad p + q = 1, \quad t_1 q s_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 15$ for 16×16 matrices and identify $\operatorname{GL}_4(R)$ with the subgroup of $\operatorname{GL}_{16}(R)$ consisting of the matrices $\operatorname{diag}(A, I_{12})$. On $M_4(R)$ consider the unital injective endomorphism

$$(7) \quad \Psi(A) = qI_4 + s_0 A t_0.$$

Here s_0At_0 denotes the matrix $(s_0a_{ij}t_0)_{ij}$. The relations $qs_0 = t_0q = 0$ and $t_0s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0\Psi(A)s_0 = A$ proves injectivity. The term qI_4 is needed for unitality because the map $A \mapsto s_0At_0$ sends I_4 to pI_4 .

Set

$$U = \begin{pmatrix} s_0I_4 & s_1t_0I_4 \\ 0 & t_1I_4 \end{pmatrix}, \quad V = \begin{pmatrix} t_0I_4 & 0 \\ s_0t_1I_4 & s_1I_4 \end{pmatrix}.$$

A direct calculation using (1) gives $UV = VU = I_8$. Put

$$(8) \quad \tau = \text{diag}(U, V) \in \text{GL}_{16}(R).$$

The Whitehead factorization of $\text{diag}(U, U^{-1})$ expresses τ as a product of elementary matrices, so $\tau \in \text{EL}_{16}(R)$.

Let $H = \text{EL}_{16}(R)$ and let $L = \text{EL}_4(R) \leq H$ be the upper-left corner. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ]. Block multiplication yields

$$(9) \quad \tau \text{diag}(A, I_{12}) \tau^{-1} = \text{diag}(\Psi(A), I_{12}).$$

For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0at_0)$. Therefore

$$(10) \quad \tau L \tau^{-1} \leq L.$$

6. THE FULL MF RADICAL

Proposition 6.1. *The group $H = \text{EL}_{16}(R)$ is simple.*

Proof. The ring $R = L_{\mathbb{F}_2}(1, 2)$ is purely infinite simple [AA] and therefore an exchange ring [Ara]. Let $N \trianglelefteq H$. Preusser's normal-subgroup theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ such that

$$\text{EL}_{16}(R, I) \leq N \leq C_{16}(R, I),$$

where $C_{16}(R, I)$ is the full congruence subgroup of level I . Since R is simple, either $I = R$ or $I = 0$. In the first case $\text{EL}_{16}(R, I) = H$, so $N = H$. In the second case

$$N \leq C_{16}(R, 0) = Z(\text{GL}_{16}(R)).$$

The center of R is $\mathbb{F}_2$ by [AC, Corollary 4.3]. For completeness, suppose that $z \in Z(\text{GL}_{16}(R))$. Commuting z with the elementary matrices $e_{ij}(1)$ shows that $z = \lambda I_{16}$ for some $\lambda \in R$. Commuting with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times = \mathbb{F}_2^\times = \{1\}$. Thus $Z(\text{GL}_{16}(R))$ is trivial, and $N = 1$ when $I = 0$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 6.2. *Let*

$$c = e_{45}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. Every elementary generator of the upper-left 4×4 corner has both indices in $\{0, 1, 2, 3\}$. The elementary commutator relations show that it commutes with $c = e_{45}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (8) gives

$$(11) \quad \tau c \tau^{-1} = e_{01}(q) e_{45}(1).$$

The second factor commutes with $\ell = e_{12}(1)$, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$. This is nontrivial because $q \neq 0$.

Let $N = \langle\langle d \rangle\rangle_H$. For arbitrary $a \in R$, two further elementary commutator identities give

$$\begin{aligned} [e_{30}(at_1), e_{02}(q)] &= e_{32}(at_1q) \in N, \\ [e_{32}(at_1q), e_{21}(s_1)] &= e_{31}(at_1qs_1) = e_{31}(a) \in N, \end{aligned}$$

where the last equality uses (6). Elementary Weyl matrices conjugate the root position $(3, 1)$ to any ordered pair of distinct indices. Thus N contains every $e_{ij}(a)$, and these elements generate H . $\square$

Proof of Theorem A. The group H is nontrivial and simple by Proposition 6.1. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ]; in particular H is finitely generated.

The containment (10), the centrality of c relative to L , and Corollary 3.3 give $d \in R_{\infty \rightarrow 2}(H)$. The subgroup $R_{\infty \rightarrow 2}(H)$ is normal, and Proposition 6.2 gives $H = \langle\langle d \rangle\rangle_H$. Hence

$$R_{\infty \rightarrow 2}(H) = H.$$

Apply Theorem B with $G = K = H$ and $D = R_{\infty \rightarrow 2}(H)$. Then $H \leq \text{Rad}_{\text{MF}}(H)$, so $\text{Rad}_{\text{MF}}(H) = H$.

If M is MF and $f: H \rightarrow M$ is a homomorphism, compose f with a faithful homomorphism from M to a norm matrix corona. Since $\text{Rad}_{\text{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion. $\square$

Corollary 6.3. *For every field k and every finite-dimensional k -vector space V , every homomorphism $H \rightarrow \text{GL}(V)$ is trivial.*

Proof. Let $\rho: H \rightarrow \text{GL}(V)$. The containment (10), the centrality of c relative to L , and Theorem 2.1 give $\rho(d) = 1$. Since $H = \langle\langle d \rangle\rangle_H$ by Proposition 6.2, the homomorphism ρ is trivial. $\square$

In particular, every homomorphism from H to a finite group, a residually finite group, or a compact Hausdorff group is trivial. For finite and residually finite targets this follows by the regular representation of a finite quotient; for compact targets it follows from Peter–Weyl point separation [PW].

7. PRESCRIBED MF QUOTIENTS

Let $A = \langle d \rangle \leq H$, where d is the element in Proposition 6.2. For a countable group Q , define

$$(12) \quad W_Q = H *_A (Q \times A),$$

where $A \rightarrow Q \times A$ is the inclusion into the second factor. The maps from H and $Q \times A$ to Q that are respectively trivial and the projection onto Q agree on A . They therefore define an epimorphism

$$\pi_Q: W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \rightarrow Q \times A \rightarrow W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (12) are injective; in particular, $d \neq 1$ in W_Q .

Proposition 7.1 (universal factorization). *Let T be a group such that every homomorphism $H \rightarrow T$ is trivial. Then precomposition with π_Q is a bijection*

$$\text{Hom}(Q, T) \longrightarrow \text{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Proof. Let $f: W_Q \rightarrow T$ be a homomorphism. Its restriction to H is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q, a) \in Q \times A$, we then have

$$f(q, a) = f(q, 1)f(1, a) = f(q, 1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q . The universal property of the amalgam shows that f factors through π_Q . The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills H , since $\langle\langle d \rangle\rangle_H = H$, and kills the second factor A in $Q \times A$. The resulting quotient is Q , with quotient map induced by π_Q . Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. $\square$

Proof of Theorem C. Every homomorphism from H to an MF group is trivial by Theorem A. Proposition 7.1 therefore gives the asserted bijection for every MF group M . In particular, every homomorphism from W_Q to a norm matrix corona factors uniquely through π_Q . Intersecting their kernels gives

$$\text{Rad}_{\text{MF}}(W_Q) = \pi_Q^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d . Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\text{Rad}_{\text{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

$\square$

The same factorization applies to every target group T for which all homomorphisms $H \rightarrow T$ are trivial. Corollary 6.3 and Peter–Weyl point separation therefore give the corresponding bijections for finite-dimensional linear groups over arbitrary fields, residually finite groups, and compact Hausdorff groups.

Corollary 7.2. *Let $N \trianglelefteq W_Q$. Then*

$$W_Q/N \text{ is MF} \iff \ker \pi_Q \leq N \text{ and } Q/\pi_Q(N) \text{ is MF.}$$

In particular, for $Q = \mathbb{Z}$,

$$W_{\mathbb{Z}}/N \text{ is MF} \iff d \in N.$$

Proof. If W_Q/N is MF, its quotient map is a homomorphism from W_Q to an MF group. Proposition 7.1 shows that it factors through π_Q , so $\ker \pi_Q \leq N$. The induced isomorphism

$$W_Q/N \cong Q/\pi_Q(N)$$

then proves the first implication. The same isomorphism proves the converse. Every quotient of $\mathbb{Z}$ is MF, and $\ker \pi_{\mathbb{Z}} = \langle\langle d \rangle\rangle_{W_{\mathbb{Z}}}$. Since N is normal, the condition $\ker \pi_{\mathbb{Z}} \leq N$ is equivalent to $d \in N$. $\square$

AI AND COMPUTATIONAL RESOURCE USAGE

AI systems were used for proof exploration, literature navigation, formal verification, and editorial revision. The authors reviewed the mathematical arguments and external references.

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